

# Bridge-Confidence Contagion in Liquid Restaking: A Pre-Registered Null Around the April 2026 KelpDAO Exploit

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## Abstract

On April 18, 2026, attackers drained 116,500 rsETH (~\$292M, 18% of supply) from KelpDAO's LayerZero bridge by forging a cross-chain message under a 1-of-1 verifier configuration. I test whether the resulting depeg in rsETH spread to other liquid restaking and liquid staking tokens that share LayerZero's Omnichain Fungible Token bridge infrastructure. Using a pre-registered difference-in-differences design on daily prices for nine LRT/LST tokens over February 3 to May 3, 2026, I find a large idiosyncratic depeg in rsETH (+781 bps post-event,  $t = 212$ ) but no statistically significant differential effect on LayerZero-bridged peers relative to non-LayerZero controls (-9.6 bps,  $t = -1.41$ ). Four pre-committed robustness specifications, including volatility, asymmetric response to ETH stress, a tight event window, and signed wedge, all return null at a Bonferroni-adjusted threshold. The result is consistent with market participants treating the event as a Kelp-specific failure rather than a LayerZero-wide bridge-protocol shock, even as protocol operators (Ethena, ether.fi) treated it as bridge-specific by precautionary pauses of their own LayerZero bridges. The divergence between operator caution and market pricing is itself the finding.

**Keywords:** liquid restaking; cross-chain bridges; LayerZero; contagion; event study; rsETH; KelpDAO

**JEL classification:** G12, G14, G19

## 1. Introduction

On April 18, 2026, at 17:35 UTC, attackers later attributed by LayerZero to the Lazarus Group's TraderTraitor subgroup (PRSol, 2026) drained approximately 116,500 rsETH (worth roughly \$292 million, or 18% of circulating supply) from KelpDAO's LayerZero-based bridge. The attack did not exploit a smart contract bug. According to Chainalysis (2026), the attackers compromised remote procedure call nodes feeding LayerZero's Decentralized Verifier Network (DVN), used a distributed denial-of-service attack to force fallback to poisoned infrastructure, and submitted a forged cross-chain message claiming Uni-Chain endpoint ID 30320 as its source. The bridge adapter was configured with a single verifier — a 1-of-1 DVN setup that Kelp has stated was the default configuration shipped at the time of its L2 expansion, and which Kelp claims 40-50% of all LayerZero OFT applications were running at the time of the exploit (Bitcoin News, 2026). KelpDAO paused contracts 46 minutes later, but the attacker had already deposited the unbacked tokens as collateral on Aave V3, Compound V3, and Euler, borrowing approximately \$236 million in WETH and other assets across Ethereum and Arbitrum (CoinDesk, 2026).

The episode triggered an immediate and visible response across decentralized finance. Aave V3 froze rsETH markets within hours, with founder Stani Kulechov confirming the exploit was external. SparkLend, Fluid, and Upshift froze rsETH positions on the same timeline. Compound V3 and Euler paused new rsETH borrows. Lido paused its earnETH product, which had a leveraged rsETH/Aave position worth approximately \$21.6 million (Crypto Times, 2026). Pendle paused its rsETH PT/YT markets. Yearn and

Beefy froze rsETH-denominated strategies. Critically, Ethena temporarily suspended all of its LayerZero Omnichain Fungible Token (OFT) bridges as a precaution despite confirming zero direct rsETH exposure; ether.fi paused LayerZero bridging for weETH and eETH; and Lombard Finance preemptively paused unrelated LayerZero LBTC routes "out of caution" (defiprime, 2026). At the protocol level, LayerZero announced on April 20 that it would no longer sign or attest messages for any application running a 1-of-1 DVN configuration, pushing a protocol-wide migration to multi-verifier setups (Bitcoin News, 2026).

DeFi total value locked fell by approximately \$13-14 billion in the 48 hours following the exploit, with Aave alone seeing between \$6.6 billion and \$8.45 billion in outflows (Bitcoin News, 2026). Some portion of these flows reflects concurrent broader risk-off conditions in crypto markets rather than direct exploit attribution. Lllamarisk projected a steady-state rsETH depeg of roughly 15.12% under uniform haircut, or 15-20% under selective recovery to L2 holders (Lllamarisk, 2026).

Two natural questions follow. First, did the rsETH depeg propagate to other LRT/LST tokens that share LayerZero's bridge infrastructure but were not directly affected, in a way that would justify the precautionary bridge pauses by Ethena and ether.fi? Second, if such contagion exists, is it specific to LayerZero, or does it appear in non-LayerZero LSTs as well, indicating a market-wide LRT risk reassessment rather than a bridge-protocol-specific shock?

I answer both questions with a difference-in-differences design pre-registered before any data was retrieved. The treated unit is rsETH. The contagion group is five LayerZero-using LRT/LST tokens (weETH, eETH, sUSDe, USDe, sfrxETH). The control group is three non-LayerZero LSTs (stETH, wstETH, cbETH). The outcome is the absolute log wedge between each token and its appropriate reference (ETH for ETH-denominated tokens, \$1 for stablecoins), demeaned by token-specific pre-event medians.

The treatment effect on rsETH is large and statistically obvious. The contagion effect on LayerZero-bridged peers is null. After token and date fixed effects, with standard errors clustered by token, the LayerZero-specific differential post-event effect is -9.6 bps ( $t = -1.41$ ), insignificant and pointing in the wrong direction. Four pre-committed robustness specifications, covering volatility, asymmetric response to ETH negative-return days, a tight five-day window, and signed wedge, all return null at Bonferroni-adjusted significance.

The finding is narrow. I do not test contagion through non-price channels (TVL flows, lending freezes, on-chain net flows), and the post-event window is approximately fifteen days including the event day. Within those limits, the result is consistent with market participants treating the Kelp event as protocol-specific rather than as evidence about LayerZero's bridge architecture broadly. Operator behavior diverged: Ethena and ether.fi paused LayerZero bridges as a precaution; the marginal trader did not reprice LayerZero-using LRTs accordingly. Either operators acted on private information that markets did not have, or operators were over-cautious relative to the price-relevant risk.

The contribution is twofold. First, this paper documents an empirical null on a question that the bridge-pause behavior of major LRT operators implicitly took as positive. Second, the design illustrates the value of pre-registration in event-study work: the headline test was committed before any post-event data was queried, and the four robustness specifications were committed in writing before any of them were estimated.

## 2. Related literature

This paper sits at the intersection of three lines of work.

First, on the economics of liquid staking derivatives. Scharnowski and Jahanshahloo (2025) provide the most direct empirical study of LST basis dynamics, documenting substantial time-variation in the stETH-ETH basis on Curve and finding that the basis is influenced by staking rewards, concentration risks, limits to arbitrage, and behavioral factors. Specifically, they show that the liquid staking basis is wider when secondary market liquidity is low, when underlying cryptocurrency returns are more volatile, and when protocol yields are low relative to direct staking. These findings motivate the choice in this paper to demean each token's wedge by its token-specific pre-event median, isolating the exploit-driven deviation from each token's normal basis level (which already reflects concentration premia, liquidity differences, and structural yield gaps that vary persistently across tokens). Xiong et al. (2024) characterize leverage staking strategies built on stETH and document widespread underperformance among liquidity providers. None of these papers covers liquid restaking tokens (LRTs), which extend LSTs by routing the underlying staked ETH through EigenLayer for additional yield, and none considers contagion across tokens after a bridge-specific shock.

Second, on cross-chain bridges and interoperability. Cao et al. (2026) provide a large-scale empirical measurement of cross-chain transfers across twenty blockchains and sixteen bridge protocols. They identify an explicit "efficiency-fragility trade-off" in which structural interoperability acts as a load balancer that reduces average gas fees but also synchronizes economic cycles, transforming independent networks into a tightly coupled system. Their analysis is at the bridge and TVL level rather than at the price level of individual bridged assets. Öz et al. (2025) measure executed cross-chain arbitrage activity across nine blockchains over a year, identifying when inventory-based versus bridge-based execution is more profitable. Zhang et al. (2026) construct a network-based fragility analysis of DeFi TVL dynamics, providing a systemic risk framework that complements bridge-specific measurement. None of these papers isolates the price response of bridged assets to a verification-layer failure of the kind documented here.

Third, on event studies of crypto market shocks and stablecoin depegs. Closer in spirit but distinct in mechanism, prior reporting on the May 2022 stETH depeg and the October 2025 USDe pricing dislocation on Binance during the 10/10 cascade (CoinGecko, 2026) treats venue-isolated or asset-isolated stress events. The Kelp event differs because the trigger is a verification-layer failure in shared bridge infrastructure, not asset-specific or venue-specific.

To my knowledge, this is the first empirical paper on the April 18, 2026 KelpDAO event. The event is too recent for any working paper to have been completed; the present paper was assembled within fifteen days of the exploit using publicly available daily price data.

### **3. Data and panel construction**

All data are daily closing prices, daily traded volume in USD, and daily market capitalization, retrieved from the CoinGecko free Demo API ([api.coingecko.com/api/v3](https://api.coingecko.com/api/v3)) on May 3, 2026. The free tier supports daily granularity for windows beyond ninety days, with a thirty-call-per-minute rate limit. No paid data is used. Replication code and the assembled panel are available on request.

The panel covers a 91-day window from February 3, 2026 to May 3, 2026. The CoinGecko free Demo API returns daily observations for windows beyond 90 days; requesting ``days=90`` returns 91 daily rows per token. Relative to the event date of April 18, 2026, this gives 74 days of pre-event data and 15 days of post-event data including the event day itself. After dropping ETH (the reference price for ETH-denominated wedges), the analysis panel contains 837 token-day observations across nine tokens.

Tokens are partitioned into three groups based on bridge infrastructure. *Treated* is rsETH alone, the directly attacked token. *LayerZero contagion* consists of five tokens that use LayerZero's Omnichain Fungible Token (OFT) standard for cross-chain movement: weETH and eETH (ether.fi), sUSDe and USDe (Ethena), and sfrxETH (Frax). LayerZero's own ecosystem documentation lists each of these as OFT integrations, and the operator response confirms it: Ethena explicitly paused all LayerZero OFT bridges after the event, and ether.fi paused weETH and eETH bridges. *Control* consists of three LSTs that do not use LayerZero: stETH and wstETH (Lido, which operates its own bridges), and cbETH (Coinbase, custodial).

The wedge variable is constructed as follows. For ETH-denominated tokens (rsETH, weETH, eETH, sfrxETH, stETH, wstETH, cbETH), the raw log ratio is  $\log(P_{token,USD}) - \log(P_{ETH,USD})$ . For USD-denominated tokens (sUSDe, USDe), the raw log ratio is  $\log(P_{token,USD})$ . Each series is then demeaned by its token-specific pre-event median, which absorbs the natural exchange-rate level (e.g., wstETH trades at roughly 1.2x ETH due to accrued staking rewards; sUSDe trades at a sustained premium to USDe). The resulting variable, expressed in basis points, measures deviation from each token's normal exchange rate to its reference.

Two design choices warrant comment. First, daily rather than hourly resolution. The CoinGecko free tier does support hourly data within the ninety-day window, but bridged-token prices on smaller venues are subject to staleness when liquidity is thin. Daily aggregation reduces this contamination at the cost of fewer observations. Second, the choice of token-specific pre-event median for demeaning rather than a fixed peg target. Some panel tokens trade persistently above or below their reference (sUSDe is yield-bearing and trades above \$1 by design); using a token-specific baseline isolates deviation from each token's normal level rather than imposing a counterfactual peg the token never targeted.

## 4. Empirical strategy

The headline specification is a difference-in-differences:

$$|wedge|_{\{i,t\}} = \beta_1 \cdot Treated\_i \cdot Post\_t + \beta_2 \cdot LZ\_i \cdot Post\_t + \alpha_i + \gamma_t + \varepsilon_{\{i,t\}}$$

where *Treated<sub>i</sub>* equals one for rsETH, *LZ<sub>i</sub>* equals one for the five LayerZero-using contagion tokens, and *Post<sub>t</sub>* equals one for dates on or after April 18, 2026. Token fixed effects  $\alpha_i$  absorb token-specific levels of normal wedge magnitude. Date fixed effects  $\gamma_t$  absorb market-wide stress days, including the broader bitcoin drawdown from \$90k to \$60k between January 28 and February 11, 2026 (VanEck, 2026), the April 1 Drift Protocol exploit, and any common LRT-class shocks not specifically tied to the Kelp event. Standard errors are clustered by token.

The coefficient of primary interest is  $\beta_2$ , the differential post-event effect on LayerZero-bridged tokens relative to non-LayerZero controls. A positive and significant  $\beta_2$  would be evidence that markets priced bridge-protocol-specific risk after the Kelp event, consistent with the operator behavior of Ethena and ether.fi. A null or negative  $\beta_2$  is evidence against bridge-protocol-specific contagion as priced by markets.

The coefficient  $\beta_1$  on the treated interaction captures the direct effect on rsETH. Its size is mechanically obvious from the public reporting of the depeg and is included as a sanity check on the wedge construction;  $\beta_1$  is not the test of interest.

Four robustness specifications were pre-committed before any robustness regression was estimated. Each tests the contagion hypothesis through a different functional form. C1 replaces the level wedge with its five-day rolling standard deviation, testing whether LayerZero-bridged tokens experience more volatile (rather than larger) wedge dynamics post-event. C2 adds a triple interaction with an indicator for daily ETH negative return, testing whether bridge risk is priced asymmetrically on bad-market days. C3 restricts the

sample to the tight  $\pm 5$  day window around April 18, where contagion would be most visible if it decays. C4 uses signed wedge instead of absolute, testing whether LayerZero-bridged tokens specifically discount (negative wedge) post-event. Bonferroni adjustment for four tests yields a significance threshold of  $p < 0.0125$ .

## 5. Results

Table 1 reports estimates from the headline difference-in-differences specification and the four pre-committed robustness checks. The treatment effect on rsETH is large and statistically obvious: 781.4 basis points,  $t = 211.8$ ,  $p < 0.001$ . The coefficient is the median post-event wedge of rsETH, expressed as a deviation from the token's pre-event median. Reported intraday spot lows during the event itself were considerably more extreme: rsETH briefly traded near \$0.71 per ETH-equivalent unit on a single venue (Phemex, 2026), implying a roughly 29% intraday depeg. The 7.8% daily-close estimate here is therefore consistent with, but smaller than, the intraday extreme: the regression measures CoinGecko's volume-weighted aggregate price at daily close, which smooths over both within-day price recovery and venue-specific dislocations. L1amarisk's projected steady-state depeg under uniform haircut is 15.12% (L1amarisk, 2026), bracketing the daily aggregate estimate from below. The wedge construction passes the sanity check.

The contagion estimate  $\beta_2$  is the test of interest. In the headline specification with token and date fixed effects and clustered standard errors,  $\beta_2 = -9.62$  bps,  $t = -1.41$ ,  $p = 0.157$ . The point estimate is small, of the wrong sign relative to the contagion hypothesis, and statistically insignificant. Dropping rsETH and estimating only on the eight non-treated tokens yields a near-identical  $\beta_2 = -9.62$ ,  $t = -1.39$ ,  $p = 0.164$ . The robustness checks fan out the test in four pre-committed directions: 5-day rolling volatility, asymmetric response to negative ETH-return days, a tight  $\pm 5$  day event window, and signed wedge. Each returns a coefficient of the wrong sign or near zero, none significant at the raw 0.05 threshold, none significant at the Bonferroni-adjusted 0.0125 threshold.

**Table 1.** *Difference-in-differences estimates*

| Specification                           | Coef.  | SE    | t-stat | p-value |
|-----------------------------------------|--------|-------|--------|---------|
| $\beta_1$ Treated $\times$ Post (rsETH) | 781.40 | 3.69  | 211.80 | < 0.001 |
| $\beta_2$ LZ $\times$ Post (contagion)  | -9.62  | 6.80  | -1.41  | 0.157   |
| $\beta_2$ contagion only (drop rsETH)   | -9.62  | 6.91  | -1.39  | 0.164   |
| C1 5-day rolling std                    | -5.51  | 5.93  | -0.93  | 0.353   |
| C2 LZ $\times$ Post $\times$ ETH-neg    | -37.49 | 30.28 | -1.24  | 0.216   |
| C3 tight window ( $\pm 5$ days)         | -4.02  | 15.23 | -0.26  | 0.792   |
| C4 signed wedge (sign matters)          | 8.25   | 16.88 | 0.49   | 0.625   |

*Notes.* Headline DiD specification:  $|\text{wedge\_bps}|_{i,t} = \beta_1 \cdot \text{Treated}_i \cdot \text{Post}_t + \beta_2 \cdot \text{LZ}_i \cdot \text{Post}_t + \alpha_i$  (token FE)  $+ \gamma_t$  (date FE)  $+ \varepsilon_{i,t}$ . Standard errors clustered by token.  $N = 837$  in the full specification (9 tokens over the 91-day window, with one token having two missing observations on the panel boundary),  $N = 744$  in the contagion-only specification (8 tokens after dropping rsETH). Robustness specifications C1 through C4 are pre-committed and reported in full; the focus coefficient is  $\beta_2$  in C1, C3, C4 and the triple interaction in C2. Bonferroni-adjusted significance threshold is  $p < 0.0125$ . No specification fires at either the raw 0.05 or the Bonferroni 0.0125 threshold.

Token-level inspection of the underlying data confirms the regression result. Among the LayerZero contagion group, post-event median absolute wedge increases for sUSDe (+22 bps), eETH (+19 bps), and

weETH (+8 bps), but decreases for USDe (-3 bps) and sfrxETH (-17 bps). Among the non-LayerZero controls, post-event medians increase for stETH (+13 bps) and cbETH (+7 bps) and slightly increase for wstETH (+3 bps). The median post-event lift is comparable across both groups (LayerZero: 8.6 bps; control: 7.7 bps), and the regression confirms the difference is statistically zero after controlling for date-specific shocks.

## 6. Discussion

Three readings of the null are consistent with the data. I cannot distinguish among them with the panel I have, and I flag each one explicitly.

**Reading one: bridge-protocol identity does not enter the marginal trader's pricing function.** On this view, market participants do not parse which bridge a given LRT uses. They reprice rsETH because rsETH is the asset with confirmed unbacked supply, but they do not extract from the Kelp event a generic signal about LayerZero. This is plausible for two related reasons. First, bridge architecture (DVN configuration, OFT adapter version) is opaque to most retail and quasi-retail participants, who dominate spot LRT trading. Second, retail price discovery runs on aggregated price feeds that smooth across venues and across the trading day, while operator decisions to pause are made on the basis of governance and risk-parameter analysis that updates much faster than retail price feeds. Öz et al. (2025) document a related asymmetry within the cross-chain arbitrage layer itself: arbitrages settled with pre-positioned inventory clear in approximately 9 seconds, whereas bridge-based arbitrages take 242 seconds. The price-correction mechanism that would propagate a bridge-specific concern across LayerZero LRT spot prices is therefore mechanically slow at exactly the bridge step that an exploit calls into question. The implied gap between operator information speed and retail price-feed speed could itself produce a null result on daily prices even if bridge-confidence risk is in fact being repriced by sophisticated participants intra-day. Testing this would require minute-level data on a venue-by-venue basis, which the daily aggregate does not support.

**Reading two: the contagion exists but is priced through non-price channels first.** Aave's freeze of rsETH lending markets, Lido's pause of earnETH, and Ethena's bridge pause all alter the option to deploy these tokens in DeFi without immediately moving spot prices. Bridge-confidence loss may show first in TVL flows, lending utilization, and protocol-level activity, with spot prices following only on a delay longer than my fifteen-day post-event window. This reading is consistent with Cao et al. (2026), who document that structural interoperability synchronizes economic cycles across coupled chains; their result implies cross-chain shock transmission that operates through TVL and capacity rather than necessarily through asset prices on each chain. Zhang et al. (2026) propose a complementary framework: a DeFi Correlation Fragility Indicator (CFI) constructed from time-varying correlation networks at the protocol category level, paired with a Risk Contribution Score (RCS) that measures the marginal contribution of different protocol types to overall systemic risk. Under their framework, the precautionary bridge pauses by Ethena, ether.fi, and Lombard can be read as observable protocol-level actions that suppress measured spot-price contagion by removing the channel through which it would propagate. The aggregate \$13-14 billion DeFi TVL drawdown in the 48 hours after the Kelp event, of which Aave alone accounted for \$6.6-8.45 billion, is consistent with this channel even though my price-based regression finds nothing. Testing the channel directly would require on-chain data on flows and utilization, which is feasible but outside the scope of a daily-price panel.

**Reading three: operator caution was over-tuned to the salient event.** Ethena and ether.fi both confirmed zero direct rsETH exposure when they paused their LayerZero OFT bridges. The pauses can be read as operator risk-management given uncertainty about the root cause, not as priced information. On this view,

the market and the operators agree on the underlying risk and the market is right that bridge-confidence risk did not materially change for non-Kelp LayerZero LRTs. The operators paused because the cost of pausing for two days is small relative to the cost of being wrong about a contagion that did not materialize.

The three readings have different policy implications. Reading one suggests that bridge architecture should be more transparent to LRT spot traders, perhaps through standardized risk disclosures. Reading two suggests that price-based event studies will systematically miss bridge-protocol contagion and that on-chain flow analysis is the right empirical lens. Reading three suggests that the operator pauses functioned as designed (a precaution that turned out to be unnecessary) and that no further response is required. The data here cannot adjudicate between them.

## 7. Limitations

**Post-event window is short.** Fifteen days of post-event data (including the event day) limits power, particularly for the tight-window robustness check. A follow-up using a six-month or twelve-month post-window would test whether contagion accumulates more slowly than spot prices reveal at the daily horizon.

**Control group is small.** Only three non-LayerZero LSTs (stETH, wstETH, cbETH) are available. The asymmetry between five LayerZero contagion tokens and three controls reflects the actual universe (Lido and Coinbase together account for most non-LayerZero LST/LRT supply) but makes the contrast statistically thinner than one would prefer.

**Pre-event window contains other crypto stress events.** The April 1, 2026 Drift Protocol exploit (\$285 million on Solana) and the broader February 2026 BTC drawdown both fall within the thirty-day pre-event window. Date fixed effects absorb their average effects but cannot isolate effects that operate differentially on LayerZero-using LRTs versus controls. The tight-window robustness check (C3) addresses this by restricting the sample to  $\pm 5$  days around April 18, and yields a similar null.

**Daily prices may smooth intraday dynamics.** Hourly contagion that decays within hours is averaged out at the daily level. The need for hourly resolution is real but the staleness contamination on smaller LRT/LST tokens at the hourly frequency was judged the larger problem at design time.

**CoinGecko prices are aggregate.** Spot prices reported by CoinGecko aggregate across venues. Venue-specific dislocations, particularly on the L2 deployments where wrapped-rsETH backing was most uncertain, may not be visible in the aggregate price.

## 8. Conclusion

On April 18, 2026, KelpDAO's LayerZero bridge was drained of \$292 million in rsETH through a forged cross-chain message, the largest decentralized finance exploit of 2026 to date. The direct effect on rsETH was a 781 basis point depeg from its pre-event median wedge. The differential effect on five other LayerZero-using LRT/LST tokens, relative to three non-LayerZero controls, was -9.6 basis points and statistically indistinguishable from zero in the headline difference-in-differences specification and in four pre-committed robustness checks. Whatever cross-token contagion the event produced did not appear in daily spot prices through the fifteen-day post-event window.

The result is consistent with operator caution diverging from market pricing. Ethena and ether.fi, both running on LayerZero OFT infrastructure, paused their own bridges as precautions, even as their tokens' spot prices showed no statistically distinguishable response from non-LayerZero peers. Whether this represents priced market efficiency, market inattention, or contagion priced through other channels (lending, TVL, on-chain flows) is unidentified by the data here. Each is a question for future work.

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